Enforcing The Chinese Wall Security Policy in Security-Enhanced Linux

by

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Abstract

The abstract is a bird's eye view of the project. It should not exceed one page. Mention the scope and objectives of the project, the methodology used,the main findings, and significance of your results.

Acknowledgements

I would like to thank

Contents

1 Introduction **1**
1.1 Background . . . 1
1.2 Project Aim . . . 1
1.3 Objectives . . . 2
1.4 Report Format . . . 2

2 Literature Review **3**
2.1 The Chinese Wall Policy . . . 3
2.1.1 The Brewer-Nash Model . . . 3
2.1.2 Intricacies of the Chinese Wall . . . 3
2.2 Chinese Walls in Adjacent Systems . . . 4
2.2.1 Chinese Wall in Unix . . . 4
2.2.2 Hypervisor-Level Chinese Walls . . . 5
2.2.3 Chinese Wall in Workflow Management . . . 6
2.2.4 Chinese Walls in Financial Institutions . . . 6
2.3 Security-Enhanced Linux . . . 7
2.3.1 Mandatory Access Control System . . . 7
2.3.2 Parts at Play . . . 8
2.3.3 Policy Module System . . . 8
2.4 Origin and Role of SELinux . . . 9
2.4.1 Red Hat Enterprise Linux . . . 9
2.5 Virtualisation of Operating Systems . . . 10

3 Specification **11**
3.1 Methodology . . . 11
3.1.1 Agile . . . 11
3.1.2 Waterfall . . . 11
3.2 Analysis . . . 11
3.3 Requirements . . . 12

4 Design **14**
4.1 Virtual SELinux Environment . . . 14
4.2 Policy Design . . . 14
4.2.1 Object Policy Modules . . . 15
4.2.2 Subject Policy Modules . . . 15
4.3 System Design . . . 15

5 Product **17**
5.1 Technology Selection . . . 17
5.1.1 Language . . . 17
5.1.2 Packages . . . 17
5.2 Implementation . . . 18

6 Verification & Validation **19**
6.1 Testing Suite . . . 19
6.2 Testing Constraints . . . 19

7 Results & Evaluation **20**
7.1 Evaluation Process . . . 20
7.2 Results of Evaluation . . . 20
7.3 Returning to the Research Questions . . . 20
7.3.1 RQ1 . . . 20
7.3.2 RQ2 . . . 20
7.3.3 Objective 1 . . . 20
7.3.4 Objective 2 . . . 20

8 Discussion & Reflection **21**
8.1 Interpreting the Results . . . 21
8.2 Reflection . . . 21
8.3 Challenges . . . 21
8.4 Limitations . . . 21
8.5 Future Work . . . 21

9 Conclusion **22**

A Appendix **24**
A.1 cwalld design . . . 24
A.2 Policy Modules . . . 26
A.3 Red Hat . . . 28

1 Introduction

1.1 Background

1.2 Project Aim

Implementation of the Chinese Wall (Brewer-Nash) Model, by leveraging the mandatory access control system introduced by Security-Enhanced Linux, is the primary goal of this project. SELinux is a well-known solution to security concerns threatening servers and personal devices. SELinux implementing Red Hat (IBM) have solidified themselves as the industry choice for secure hosting solutions, being a leading operating system used by over 90% of Fortune 500 companies for its quality privacy guarantee [2].

The Chinese Wall's dynamic nature makes it an intriguing security policy, which let it stand in stark contrast from typical static security policies. For this reason or others, it has received a noticeable lack of attention in the space surrounding SELinux. While providing an implementation for SELinux, this report delves into the use cases of this specific policy, where it has succeeded in the past, and what its inclusion in SELinux allows.

"The more you know, the less you can share" serves the basis for the Chinese Wall, where a subject (in our context: a process, daemon or user) in the system loses more privileges as it completes its lifecycle. Conflict of Interest (`CoI`) classes dictate the main restrictions, preventing the subject from sharing sensitive information between objects (files) that are conflicting with each-other. A lattice based system is used to transition subject labels based on their read and write history accordingly, which should suffice to handle all cases of the Chinese Wall.

Previous research into both Chinese Wall effectiveness and SELinux was factored in the design process, along with hands on experimentation and prototyping of different solutions. The development of the Chinese Wall daemon should serve to open possibilities of dynamic policies in the otherwise static SELinux, and potentially lead the way in future security solutions.

To enforce this policy model in SELinux, policy modules must be compiled into the kernel via the security server and work in tandem with the designed Chinese Wall Security Daemon, developed to be capable of enforce dynamic label alterations to monitored subjects in a directory. Finally, this allows the SELinux Access Vector Cache (AVC) to handle the restrictions on a kernel level, successfully locking the system down to `CoI` breaches.

1.3 Objectives

1.1 Create a suitable virtual environment for SELinux.

1.2 Understand the principles of:

1.2.A The Chinese Wall Security Model.

1.2.B SELinux's security server & type enforcement system.

1.3 Implement a static implementation of the Chinese Wall policy, confined by SELinux.

1.4 Develop a system daemon capable of monitoring subjects and their operations on objects.

1.5 Update daemon to dynamically alter labels, allowing SELinux to enforce the Chinese Wall policy.

1.6 Fully integrate the Chinese Wall Daemon into SELinux's environment, complete with its own policy module.

1.7 Test and analyse the capabilities and limitations of the Chinese Wall Daemon.

1.8 Analyse the difficulties that arose with development and conclude why development in this area has been lacklustre.

1.4 Report Format

The report is structured in an sequence appropriate to the process undertook addressing this challenge, detailing background research, solution design, implementation and testing, followed by results and evaluation.

2 Literature Review

2.1 The Chinese Wall Policy

2.1.1 The Brewer-Nash Model

The Chinese Wall Model - or Brewer-Nash Model - is a security model that aims to prevent information flow between datasets that are in *conflicts of interest* (`CoI`) [5]. The Brewer-Nash Model provides a mathematical format that is based on the larger Chinese Wall policy. As opposed to the classic multi-level security Bell LaPadula model, which enforces *"no read up, no write down"* [4], the Chinese Wall model allows for information flow freely until a subject attempts to read from a `CoI` object. Upon this access, the system shall dynamically revoke their write permissions to other objects in the `CoI` group, creating a "wall" in-between them. Consider the following scenario:

A data server stores information on Bank A and B, and Tech Company A and B. These are in a `CoI` relation with their respective counterparts. If a user approaches the dataset, they are initially able to access any object. As far as the system is concerned, this user is free as they have no prior knowledge of either companies. Suppose the user has read from Bank A, the situation develops - the user now possesses information that, deemed by the Chinese Wall, should not be allowed to reach Bank B, as they in a `CoI` relation with Bank A. The Wall revokes the write access from the user in regard to Bank B. Now information *cannot* flow from Bank A to B. The system is constructed to definitively prevent this leakage scenario. It should also be noted that once information from Bank A is read, it cannot be shared to Tech Company A or B. In that scenario, a different subject could read from Tech Company A and write to Bank B, as they are not in conflict, exploiting the rules of the Chinese Wall.

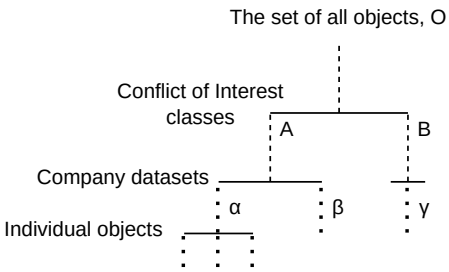


Figure 1: The composition of objects

2.1.2 Intricacies of the Chinese Wall

Brewer & Nash's rules demonstrated the Chinese Wall as more as a mathematical concept than a policy. They broke it down into the following rules, influenced by Bell-LaPadula:

- **Simple Security Property**: Subject `S` can read object `O` only if `O` is in the same Company

Dataset (CD) as an object already read by S, or if O's CD is in a different CoI group from all CDs of objects already read by S.

- ***-Property**: Subject S can write to an object O only if S can read O **and** all objects O' read by S are in the same CD as O.

The Brewer-Nash model suffers from a lack of clarity in some areas, leaving readers split on interpretations of certain definitions in the paper. The **-property* for one, leaves it unclear if write access implies read access. As such, this report will outline a clear understanding the Chinese Wall and follow that definition.

After the Brewer-Nash model was communicated, another paper was published once again by Brewer & Nash, but also T.Y. Lin, who presented them with potential improvements on their model after their original paper, leading to the *Aggressive Chinese Wall* [11]. This idea, trying to disprove the assumption that the conflict of interest is an equivalent relation, has since been accepted into the mainstream.

2.2 Chinese Walls in Adjacent Systems

2.2.1 Chinese Wall in Unix

A Chinese Wall model implementation was made in standard Unix, leveraging its *Discretionary Access Control* (DAC) security system also adopted by Linux and macOS. The 1997 Unix proof by S.N. Foley built upon T.Y. Lin's Aggressive Chinese Wall Model, illustrating the possibility of its creation in with DAC [7]. The Unix proof features a CoI matrix, detailing a non-transitive version of the CoI relations. Foley's concept focused on the restriction of users, preventing a user of the computer from spreading information enforced via their user ID and group ID. DAC is defined such that objects and their privileges are managed by their creator, and the super-user can access and alter anything. DAC enforces based on the user ID and any groups with which they belong. While DAC brings about great flexibility, the models' reliance on ownership often leads to uncontrolled privilege propagation. It is known to be difficult to measure the security of the DAC model, and in some cases it is undecidable [6].

2.2.2 Hypervisor-Level Chinese Walls

Chinese Walls have been implemented in Cloud Computing against various threat models. Cloud Computing involves many Virtual Machines (VMs) running on one core system, with the virtual machine monitor, or *"hypervisor"* acting as the delegating force of the computers' hardware to the VMs. Each VM is a separate Operating System (OS), meant to be completely independent of the other VMs on a machine.

sHype Hypervisor

The sHype *Mandatory Access Control* system was added to the Xen hypervisor to defend against information sharing between VMs by R. Sailer et al [13]. The hypervisor prevents inter-VM communication with kernel hooks, intercepting and evaluating every attempt to communicate across VM's. Whilst claiming to be a Chinese Wall, the sHype architecture does not truly capture the notion of Brewer & Nash's Model, instead focusing on static *"Chinese Wall types"*. Regardless, sHype does successfully prevent against `CoI` leakages, in the case of insider activities or inter-VM attacks.

Chinese Wall Central Management System

Inter-VM attacks can happen if a VM on the system is breached, and can allow a targeted attack on another VM, even a compromised hypervisor. To eliminate the possibility of inter-VM attacks from competitors, a different Chinese Wall security model was implemented. The CWCMS, operating above the *Kernel-based Virtual Machine*[1] level via installed control agents, with a sole focus of actively preventing `CoI` breaches was designed by Tsai et al [15]. Once a VM in a `CoI` group is running on a physical host, the Wall prevents other VMs belonging to the same `CoI` group from being scheduled on the same machine in advance. This is a pre-arranged decision made by graph colouring, as opposed to sHype which simply attempts to deploy it on different systems until one is satisfactory. This effectively eliminates the risk of inter-VM attacks in these systems. Similar to sHype, CWCMS also lacks dynamic label profiling, resolving to a statically enforced, yet successful, policy model.

[1] KVM allows the Linux kernel to act as a hypervisor.

Security-Awareness VMs Management Scheme

Finally, a dynamic Wall was created in Yu et al's SVMS system [16]. SVMS aims to eliminate intrahost *side channel attacks*, where malicious users extract private information by reading from shared data caches on the host machine. Yu et al address this with two techniques: Preventing simultaneous execution, and a more unique advance with VM placement and migration solutions based on VM traces. These VM traces are accumulated by tracking the *aggressive conflict of interest relation* and *aggressive in ally with relation*, they create a dynamic policy model faithful to the Brewer-Nash model, proving that the rules meet the simple-security property and *-security property.

All models, while different, agree that the *"Chinese Wall"* for Cloud involves preventing conflicting resources from running at the same time. Whilst not exactly cohering to the Brewer-Nash model, they all manage to succeed in their endeavours to create a Chinese Wall.

2.2.3 Chinese Wall in Workflow Management

Beyond virtualized environments, the Chinese Wall model has also successfully been implemented in workflow management systems. *Workflow Management Systems* (WfMS) are intrinsically network-based applications, potentially hosting multiple conflicting company datasets. As such, a secure WfMS must implement security features, specifically targeting information confidentiality as a main point of effort. The Chinese Wall security model can be leveraged to prevent conflict of interest leakage in this environment. The authors proposed and implemented a *Chinese Wall Security Policy Specification Language* in Java [9]. The paper mentions that *Role Based Access Control* (RBAC) is unable to implement general cases of the Chinese Wall, as it has issues implementing the Brewer-Nash simple security property in dynamic contexts. The dynamic nature of the Chinese Wall is proven to be difficult to implement with standard tradition.

2.2.4 Chinese Walls in Financial Institutions

Chinese Walls serve as a pillar of compliance in the financial sector. The ethical policy, more commonly just referred as *The Wall* in finance, is a commonplace measure that should theoretically and practically prevent misuse of information leakage in banks [8] and other financial management firms. Thus, the Wall is more attuned to person-level restrictions as opposed to system-wide enforcement as is present in modern considerations of the Wall. While entirely different from modern

systems, it helps understand why the Wall is present in different systems

2.3 Security-Enhanced Linux

2.3.1 Mandatory Access Control System

Security Enhanced Linux implements a *Mandatory Access Control* (MAC) system, providing a policy-driven security layer extended beyond the classic DAC. Utilisation of labels - applied to both objects and subjects, and verbose policy modules provide the OS great freedom in restriction and enforcement. The *Security Context*, or *label* of an object/subject takes the form `user_u:role_r:type_t:s0` and can be viewed with the *-Z* flag when using *ls* or *ps*[2].

```
/home/testgrounds/objects ❯ ls -Zl | grep alpha_logs
-rw-rw-rw-. 1 cal cal unconfined_u:object_r:alpha_t:s0     28 Mar 20 00:24 alpha_logs
      Standard Unix DAC            SELinux MAC Security Context
```

Figure 2: Security Context of an object

The Security Context consists of:

- **User** (`user_u`): Used to give the confine speficic Linux users to a class, such as guest_u, user_u or staff_u.
- **Role** (`role_r`): Used to confine users to roles, implementing role based access control.
- **Type** (`type_t`): Used to restrict subjects based on their domain type. The most specific field a policy can target.
- **Level** (`s0`): Used to enforce classes between types. It can be used to seperate VM's and such from sharing types in a *Multi-Level Security* context.

For instance, a simple policy may be that an object of type `httpd_t` can only be operated on by a subject of type `httpd_t` and not `sshd_t`. This provides strong mitigation against broken access control damages, as subjects can only access objects that they are contained to by SELinux's security server. In a case where a subject gets hijacked, it is still only restricted to the domain it originally had on the system.

[2] *ls* and *ps* are standard Unix commands to list objects and processes respectively.

2.3.2 Parts at Play

Being based off of the Flux Advanced Security Kernel (FLASK) security architecture for flexible nondiscretionary access controls, the architecture of SELinux is split into 2 main bodies:

- Policy enforcement via the Access Vector Cache (AVC)
- Policy decision-making via the policy repository and security server

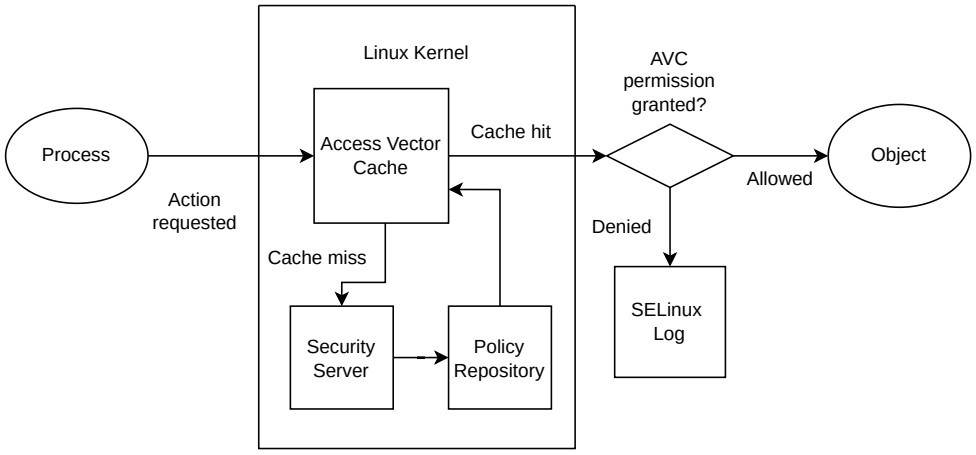


Figure 3: SELinux Components

2.3.3 Policy Module System

The *policy modules* (PMs) that SELinux provides are solely controlled by the super-user. Red Hat 9 by default has **440** modules, ranging every domain in the system. The *policycoreutils* package introduces developer tools to install and alter PMs on the system, calling *sepolicy generate –init* from the command line generates boilerplate for the policy including:

- Type Enforcement *(.te)* file: Unique syntax notates dependencies and designs rules, introduced to restrict the new labels.
- File Contents *(.fc)* file: Details the file contents related to the policy.
- RPM Spec *(.spec)* file: Installs the policy into the kernel, including labelling, the interface file and associated man page of the policy.
- Interface *(.if)* file: Defines the interfaces for the types generated in the type enforcement file, allowing policies to reference each-other.

- Shell Script *(.sh)* file: A script to be run by the super-user to compile and install the policy module. Under the hood, various commands are executed, eventually calling *semodule -i* to complete installation.

Once installed, PMs are persistent, and restrict based on their domain while SELinux is in *Enforcing* mode. SElinux also supports a *Permissive* mode, where denials are simply logged and not enforced, useful for testing at the cost of risk in the system. The *sepolicy generate* command includes a *permissive* line that must be removed from each *.te* file for the same reason.

2.4 Origin and Role of SELinux

SELinux is adopted by many distributions of Linux, including Red Hat Enterprise Linux (RHEL), Fedora and even mobile operating system giant Android. The origin of SELinux can be attributed to the *National Security Agency* (NSA)'s workings on a series of books called the Rainbow books - A series of Computer Security standards and guidelines by the United States Government. They published one and created a second (unpublished) book that included a formal SELinux model and associated evaluation evidence prototype. SELinux was a demonstration that mandatory access control was viable and could be added to the Linux kernel. The NSA, along with NAI Labs were responsibile for using FLASK to integrate MAC controls into the Linux kernel [14][12].

Linux, as an OS was able to adopt this because of its commitment to software freedom and ethical use of technology, originally being an open-source derivation from the Unix that was locked down by AT&T's IP. Linus Torvalds combined his Linux kernel with the GNU system tools by the GNU Project (Richard Stallman and various developers) to create GNU/Linux. This open-source philosophy was a key factor enabling SELinux's integration into the mainline kernel in the 2.6 series in December 2004.

2.4.1 Red Hat Enterprise Linux

The impact of Linux is huge, with 85% of smartphones [10] and 96% of top 1 million websites [1] powered by the free OS. Tech Giant IBM produced RHEL has a 33% market share in worldwide operating systems [3], no doubt because of their commitment to reliability, stability and security.

As one of many distributions of Linux, RHEL is a unique as a subscription based distribution, with

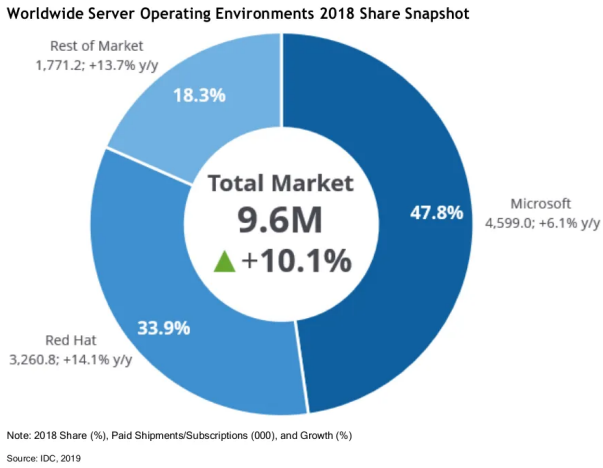


Figure 4: Red Hat statistics

a target audience of companies expecting nothing less than the finest. This promotes SELinux as a lucrative avenue for development.

2.5 Virtualisation of Operating Systems

As this project involves a virtualised Red Hat environment

This section should provide the grounding for your project. You will be able to refer back to this in Section 3, where you report on design decisions and methodology. There should be a strong link to your research questions (for project types 2 and 3). There will be a strong link to the software design decisions for project type 1. Don't just include a paragraph for each paper you read. Synthesize it and create a story line. Review between 5 and 10 authoritative sources. For research based projects, some of these should be from peer-reviewed journals or conferences. This section will reference literature like this. Sloppy referencing will (1) take time and effort, and (2) lose you marks if you don't do it.

3 Specification

3.1 Methodology

The decision to adhere to a methodology early in development helps set up consistent and satisfying progress over the life of the project. As this project was undertaken as a solo developer, the decision of methodology was not as crucial as a team setting. Still, the project lifecycle followed is more closely attuned to the agile principle.

3.1.1 Agile

Agile methodology focuses on multiple iterative stages, constantly evaluating the progress and direction of the project, and tuning decisions to reach the solution. As SELinux is a system with predefined capabilities and limitations, various different methods were used throughout to understand the boundaries and prototyping was updated accordingly. Experimentation in SELinux was carried out simultaneously alongside research, allowing accumulated insight to be backed by personal experience, and a more complete and satisfying solution.

This agile process was accompanied by a simple kanban board, easily conveying the stage of the project as well as next steps and currently occupied tasks. This allowed for a more streamlined lifecycle of an idea.

3.1.2 Waterfall

Waterfall involves a predefined set of requirements, of which progress is made solely to the final objective. This methodology benefits experience, where familiarity with the project environment allows decisions to be made far in advance. Waterfall usually has issues with bugs, as testing is only a focus in the later stages of development. As this is a project within time constraints and a single developer, agile was selected.

3.2 Analysis

To enforce a dynamic security policy, its natural to deduce that some sort of process must be constantly running to be capable of enforcing said dynamism. During experimentation on the boundaries of SELinux, it was surmised that the most effective way to leverage its MAC system was to change the type field of the security context on both subjects and objects. The user, role and

level fields of the subjects/objects were not sufficient to enforce a security policy, as policy modules were unable to be compiled to enforce restrictions based entirely on said fields.

Continuous experimentation on the type field of the label preceded the philosophy that would be the main premise of the Chinese Wall Daemon. Pre-compiled policy modules for all scenarios, and a daemon that altered the labels of subjects at runtime according to their behaviour and the Chinese Wall policy. This daemon would also need to be fully integrated into SELinux's system, running under its own `cwalld_t` type rather than `unconfined_t`.

Policy modules are unable to be added to the SELinux security server at runtime, as it requires a complete re-compile of the security server and kernel. This operation is capable of taking anywhere between 4 and 8 seconds - unrealistic at runtime for a background process.

3.3 Requirements

Requirements specified early in the agile cycle follow the MoSCoW principle. Requirements are split into 4 categories:

- Must have - features that must be implemented to achieve the goal.
- Should have - features that should be implemented to develop the solution.
- Could have - features that could be implemented but are not priorised.
- Won't have - features that were considered but decided against.

Functional requirements:

Non-functional requirements:

These requirements were solidified after considering the functionality that would be needed to enforce the Chinese Wall. While the requirements were decided from the beginning, the means of which to complete them was left ambiguous, in line with agile structure.

Code	Requirement	MoSCoW
FR1	`cwalld` must alter necessary configurations in the system before running.	**Must**
FR2	`cwalld` must monitor all operations made by subjects across a given directory.	**Must**
FR3	`cwalld` must make decisions based on subject label and object label aligned with the Chinese Wall policy.	**Must**
FR4	`cwalld` must alter the labels of subjects at upon call.	**Must**
FR5	`cwalld` must be accompanied by policy modules for objects.	**Must**
FR6	`cwalld` must be accompanied by policy modules for subjects.	**Must**
FR7	`cwalld` should be confined by SELinux with its own custom policy.	**Should**
FR8	`cwalld` should run as a service in the system with its own service declaration file.	**Should**
FR9	`cwalld` should write all operations to a log.	**Should**
FR10	`cwalld` should include a binary to follow and display the log in real time.	**Should**
FR11	`cwalld` could automatically create necessary policy modules for its runtime.	**Could**
FR12	`cwalld` could automatically install necessary policy modules before running.	**Could**
FR13	`cwalld` could dynamically enforce the policy given a set of `CoI` classes as input.	**Could**
FR14	`cwalld` won't enforce the Chinese Wall policy on the system-wide.	**Won't**

Code	Requirement	MoSCoW
NFR1	`cwalld` shall solely be controlled by the super-user.	**Must**
NFR2	`cwalld` shall run as a daemon in user-space of the computer.	**Should**
NFR3	`cwalld` shall be written with structural, intentional and idiomatic code.	**Should**
NFR4	`cwalld` shall be a good UNIX citizen, following classic system conventions.	**Should**
NFR5	`cwalld` shall function as intended across both x86 and ARM architecture.	**Should**
NFR6	`cwalld` shall act atomically, ensuring the Chinese Wall is instantaneous.	**Could**

4 Design

Describe all details of the design and procedures used to achieve the project objectives. Do this chronologically.

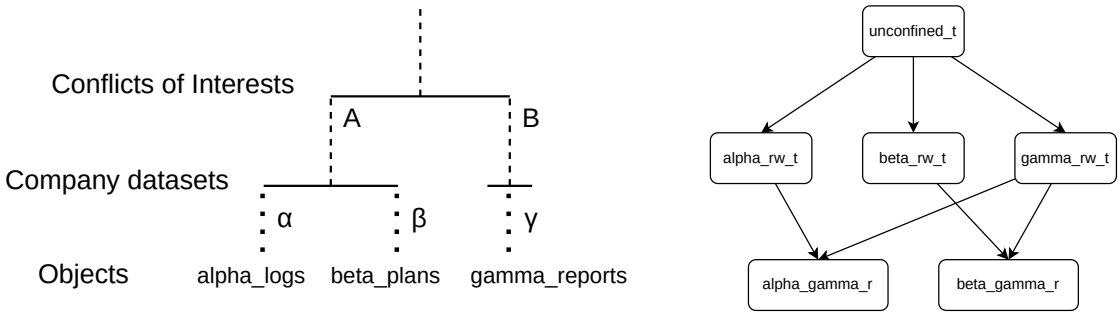
4.1 Virtual SELinux Environment

To develop and test on SELinux, the required environment is necessary. RHEL was chosen, as it is the most commonly used SELinux distribution. Fedora was also considered; a popular fork of Red Hat skewed towards personal workstations rather than enterprise servers. As RHEL is an enterprise product, the Red Hat Developer program had to be infiltrated to gain access for free. Once a key was acquired, a virtual machine was created using KVM, *Quick Emulator* (QEMU) and *Virt Manager* (IBM) to interface with the virtual machine.

Development was split between the Virt Manager interface and an *ssh reverse tunnel*, allowing the virtual machine to be navigated from the command line of the host machine. RHEL 9 and RHEL 10 were both used for experimentation, RHEL 9 being used after the instance of RHEL 10 was bricked during testing.

4.2 Policy Design

To demonstrate the Chinese Wall Model in SELinux, the minimum amount of labels necessary to capture all use cases of the wall was designed. The model was designed such that there would be 3 object classes, 2 of which were in a `CoI` relation, and one of which was uninvolved.



(a) Object structure (b) Subject label transition lattice

Figure 5: SELinux Chinese Wall

4.2.1 Object Policy Modules

A unique PM had to be installed for each different `CoI` type. The object PMs were simple, as they only needed to introduce the type to the system. This allowed the subject PMs to be reference the objects. The following types were to be introduced:

- `alpha_t`
- `beta_t`
- `gamma_t`
- `sanitised_t`

4.2.2 Subject Policy Modules

By default, SELinux restricts all actions that aren't allowed in some PM. Thus, design of subject PMs must explicitly allow the object operations they shall allow. Six labels are included in the design.

Label	`alpha_t`	`beta_t`	`gamma_t`	`sanitised_t`
`unconfined_t`	r/w	r/w	r/w	r/w
`alpha_rw_t`	r/w		r	r
`beta_rw_t`		r/w	r	r
`gamma_rw_t`	r	r	r/w	r
`alpha_gamma_r_t`	r		r	r
`beta_gamma_r_t`		r	r	r

These labels may be unconventional, for instance, why is it that `alpha_rw_t` can read from `gamma_t`? SELinux is a preventative measure, where standard Wall method of a subject *asking* for permission to access an object or subject is just not possible. As such, the enforcement involves allowing the subject to access anything that it *could*, instantly altering its label afterwards. When a subject with label `alpha_rw_t` *can* reads from `gamma_t`, its label instantly transitions to `alpha_gamma_r_t`, revoking its write access to `alpha_t` forever.

4.3 System Design

To assist in development, a UML diagram was created:

Show how you designed your database (if appropriate) and how you designed your system architecture, and the individual parts. Use UML and an Entity Relationship diagram It should be detailed enough to allow for an assessment of the rigour of your process, and, in the case of research projects, in terms of how well grounded your research is in the research literature. In these cases, refer back to relevant sections in the previous chapter. Say which

software lifecycle approach you used e.g., Waterfall, Spiral, Agile. How did you gather user requirements?

5 Product

5.1 Technology Selection

5.1.1 Language

Go *(or Golang)* was chosen for the development of the Chinese Wall Daemon, due to its strong affinity with system-level programming. Go's advantages in systems engineering range from such:

- **Systems Origin:** Go was developed with systems programming at its core, remaining unpunishing while giving the developer great versatility and library access.
- **Performance:** As a compiled language, the flexibility to be built into a binary means minimal overhead at runtime. Go promises low-latency, high-performance applications.
- **Modernity:** Go is a modern alternative to the otherwise stagnant systems engineering environment. The focus on clean, idiomatic code with statically enforced typing means debugging can be trivial if common practice is followed.
- **Familiarity:** The simplicity of Go discourages unnecessary complexity while promoting hands-on learning. Repeated experience working with the language supported the choice and guaranteed steady progress in the venture.

5.1.2 Packages

nxadm/tail

A fork of the deprecated tail package by *hpcloud*, *nxadm* provides a Tail object that can be used to tail a specified file. Tailing a file involves effectively attaching a process to a file (typically a log file), and returning updates in real time. It allows a program to closely monitor a file for its own needs. The package leverages *fsnotify* to instantly recognise file updates, also providing truncation/move support, allowing it to follow rotating log files with minimal configuration.

opencontainers/selinux

The *selinux* package provides a direct route for Go make SELinux system calls to the kernel, necessary for operations such as *chcon* (change context, i.e label of an object) and the *-Z* flag for acquiring label information. It also provides native error handling to Go, streamlining usage and debugging.

x/sys/unix

The official Go Project *sys/unix* package introduces a layer of abstraction over both *x86* and *ARM* architecture system call codes, providing a unified interface for detecting system operations.

coreos/go-systemd

go-systemd introduces tools to manage a connection directly to the system *D-Bus*. This allows direct communication with operations typically managed with *systemctl*, such as starting and stopping services on a system.

5.2 Implementation

Provide implementation details: language used, architecture (e.g. server and client, or hub and spoke). Explain how you secured personal details.

6 Verification & Validation

6.1 Testing Suite

6.2 Testing Constraints

How did you verify that your software was debugged and worked correctly? How did you validate that you had implemented all the functional and non-functional requirements?

7 Results & Evaluation

Results are often presented in tables, figures and other relevant illustrations. Include text that refers to these figures/tables.

7.1 Evaluation Process

If you involved humans in the evaluation, how many did you have? What can you say about the demographics of your participants? (If you did collect these) In terms of the user interface, how did you carry out a usability evaluation, how did you go about doing this? How did you recruit participants? In terms of delivering functionality, did you carry out user acceptance testing? (see attached guidance).

7.2 Results of Evaluation

This Section includes a direct interpretation of the gathered data and evaluation processes.

7.3 Returning to the Research Questions

Return to research questions or objectives as appropriate.

7.3.1 RQ1

It is clear from our findings that James Bond was born in Wigtown in Scotland. However, he grew up in Diss, in Norfolk. We know this because ....

7.3.2 RQ2

We were not able to answer this question from our studies, although some suggestions were made. These could not be proven.

7.3.3 Objective 1

7.3.4 Objective 2

8 Discussion & Reflection

8.1 Interpreting the Results

Here you will discuss your findings. This is especially relevant for research projects. You might interpret what the data and evaluation implies, both for future research and for practice (if appropriate). The discussion is not a review of literature. You should try to compare research findings with previous work, provide explanations for your findings, discuss research findings, in terms of their contribution.

8.2 Reflection

Look back and think about what you would do differently if you were going to start the project with the knowledge you have now. Be honest about your mistakes or missteps.

8.3 Challenges

This is not the place to mention personal circumstances but rather challenges related to the work involved in the project.

8.4 Limitations

Acknowledge things like: small number of participants, software wasn't completely debugged, or whatever else went wrong and affected your project. Include as appropriate

8.5 Future Work

If someone else wanted to build on your project's product, what would be cool to do next?

9 Conclusion

The conclusion is similar to when a plane lands. You don't rewrite the introduction. You say something like - I addressed the problem outlined in the introduction, and I built some software to do this. I tested the software like this ADD FEW WORDS. Summarize main findings drawn from the project work. Mention the objectives or research questions. Do not repeat points raised in the discussion and reflection Section. If applicable, you can make recommendations. The conclusion should NOT contain any references.

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A Appendix

A.1 cwalld design

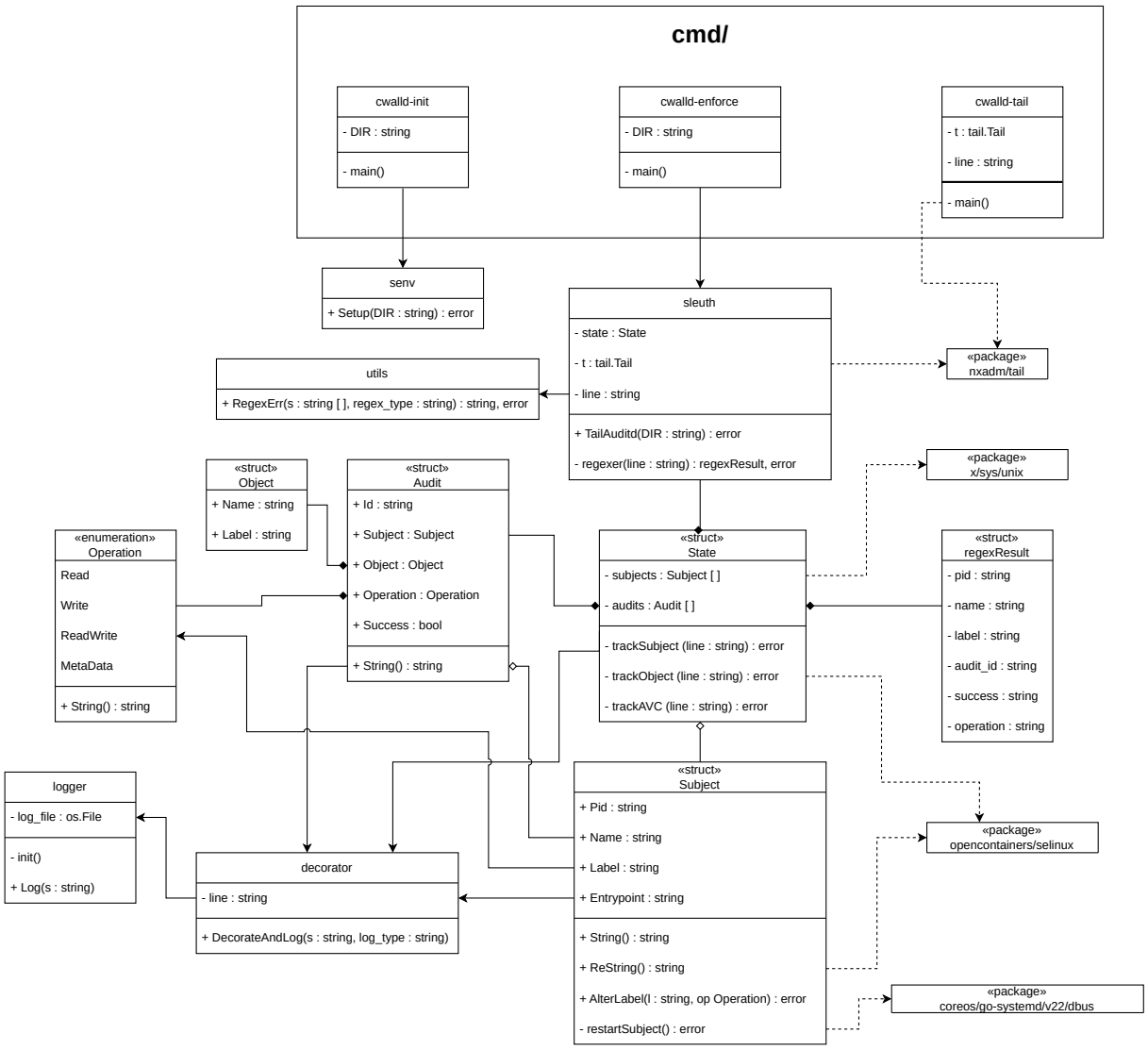

Figure 6: UML diagram

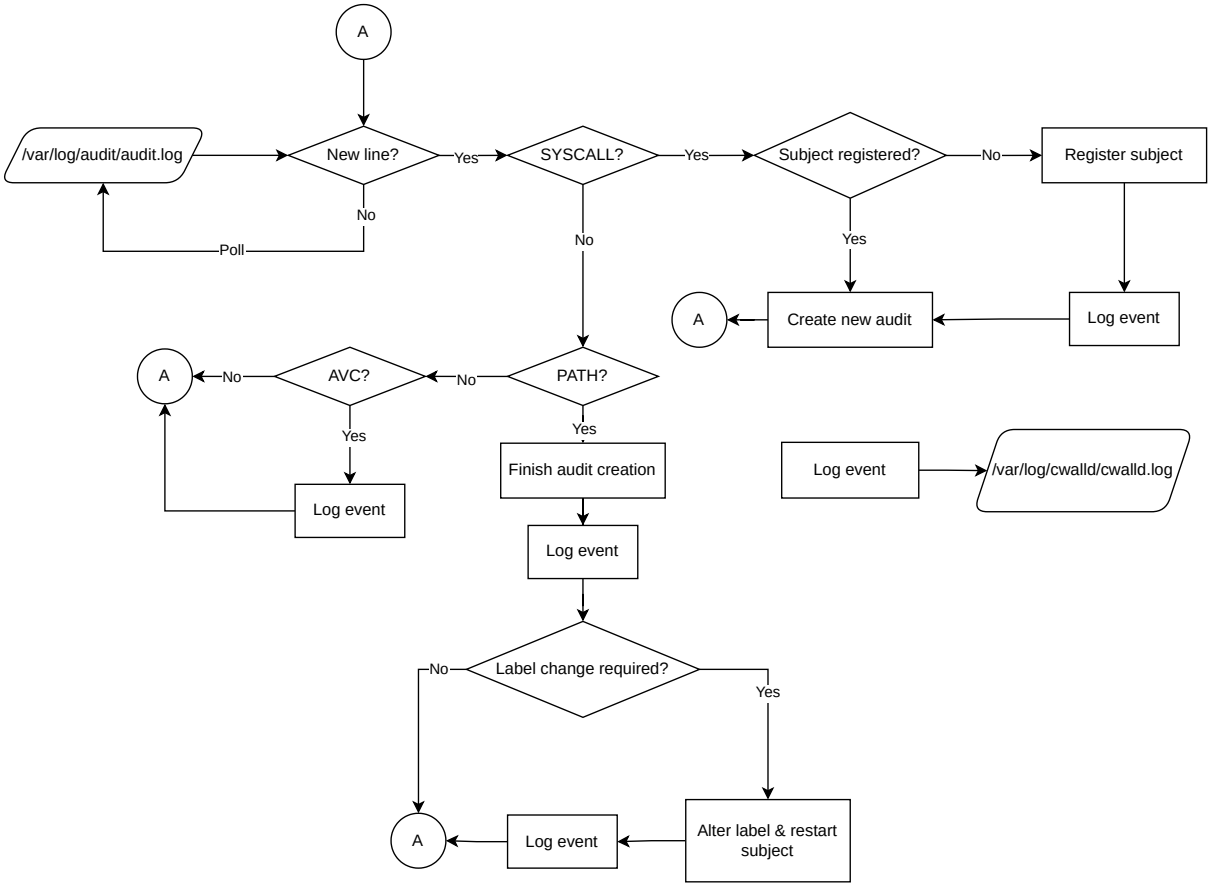

Figure 7: Flowchart

A.2 Policy Modules

```
policy_module(alpha, 1.0.0)

########################################
#
# Declarations
#

type alpha_t;
type alpha_exec_t;
init_daemon_domain(alpha_t, alpha_exec_t)

########################################
#
# alpha local policy
#
allow alpha_t self:fifo_file rw_fifo_file_perms;
allow alpha_t self:unix_stream_socket create_stream_socket_perms;

files_type(alpha_t)

domain_use_interactive_fds(alpha_t)

files_read_etc_files(alpha_t)

miscfiles_read_localization(alpha_t)
```

Figure 8: *alpha.te* file

```
policy_module(alpha_rw, 1.0.0)

########################################
#
# Declarations
#

type alpha_rw_t;
type alpha_rw_exec_t;

require {
	type alpha_t;
	type sanitised_t;
	type user_home_t;
	type gamma_t;
}

init_daemon_domain(alpha_rw_t, alpha_rw_exec_t)

########################################
#
# alpha_rw local policy
#
allow alpha_rw_t self:fifo_file rw_fifo_file_perms;
allow alpha_rw_t self:unix_stream_socket create_stream_socket_perms;

domain_use_interactive_fds(alpha_rw_t)

files_read_etc_files(alpha_rw_t)

miscfiles_read_localization(alpha_rw_t)

# cwalld policies
allow alpha_rw_t alpha_t:file { read write getattr open };
allow alpha_rw_t sanitised_t:file { read getattr open };
allow alpha_rw_t gamma_t:file { read getattr open };

# navigational policies
allow alpha_rw_t user_home_t:dir { search read open write add_name };

```

Figure 9: *alpha_rw_t .te* file

A.3 Red Hat

```
           .MMM..:MMMMMMM                    root@localhost
         MMMMMMMMMMMMMMMMMMM                 --------------
         MMMMMMMMMMMMMMMMMMMM.               OS: Red Hat Enterprise Linux 9.7 (Plow) x86_64
        MMMMMMMMMMMMMMMMMMMMMM               Host: KVM/QEMU Standard PC (Q35 + ICH9, 2009) (pc-q35-10.2)
       ,MMMMMMMMMMMMMMMMMMMMMM:              Kernel: Linux 5.14.0-611.41.1.el9_7.x86_64
       MMMMMMMMMMMMMMMMMMMMMMMM              Uptime: 1 min
  .MMMM'  MMMMMMMMMMMMMMMMMMMMMM             Packages: 1398 (rpm)
 MMMMMM    `MMMMMMMMMMMMMMMMMMMM.            Shell: bash 5.1.8
MMMMMMMM       MMMMMMMMMMMMMMMMMM .          Display (Virtual-1): 2560x1600 @ 60 Hz
MMMMMMMMM.        `MMMMMMMMMMMMM' MM.        WM: Mutter (X11)
MMMMMMMMMMM.                       MMMM      Cursor: Adwaita
`MMMMMMMMMMMM.                   ,MMMMM.     Terminal: GNOME Terminal 3.40.3
 `MMMMMMMMMMMMMMM.             ,MMMMMMMM.    Terminal Font: Source Code Pro (10pt)
    MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM  CPU: 4 x 11th Gen Intel(R) Core(TM) i7-11390H (4) @ 3.42 GHz
      MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM:  GPU: Red Hat, Inc. QXL paravirtual graphic card
         MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM   Memory: 1.72 GiB / 5.53 GiB (31%)
            `MMMMMMMMMMMMMMMMMMMMMMMMM:     Swap: 0 B / 2.00 GiB (0%)
                ``MMMMMMMMMMMMMMMMMM'       Disk (/): 10.70 GiB / 16.93 GiB (63%) - xfs
                                            Local IP (enp1s0): 10.0.2.15/24
                                            Locale: en_GB.UTF-8
```

Figure 10: Red Hat system specifications